SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1506

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 40**

Question Count: 5

Duration :60 Minutes

Code of Conduct

I **Dinesh V (192524309)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **Dinesh V**

Assessment Tasks

Industry Problem Solving Task

1. Retail Data Optimization Problem

A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

Answer:

A unified big data architecture should integrate data from physical stores, the online platform, inventory systems and customer interactions.

• Data sources: POS systems, e-commerce transactions, inventory databases, customer profiles, website/mobile activity, loyalty programs and supplier data.

• Ingestion: Apache Kafka can capture real-time events, while batch ingestion handles historical data.

• Storage: A cloud data lake such as Amazon S3, Azure Data Lake Storage or Google Cloud Storage can store raw data. A cloud data warehouse can store cleaned structured data.

• Processing: Apache Spark can perform distributed batch and streaming processing.
• Real-time inventory: Streaming jobs update stock levels after sales, returns and stock receipts.
• Demand forecasting: Historical sales, seasonality, promotions, holidays and price data can be used with regression, Random Forest, Gradient Boosting or time-series models.
• Recommendations: Collaborative filtering, content-based filtering or hybrid recommendation models can use purchase and browsing behaviour.
• Serving: APIs and low-latency databases can deliver current inventory and recommendations to applications.
• Security: Use encryption, authentication, role-based access control, auditing, privacy controls and data-quality validation.

This architecture reduces data silos and supports real-time inventory visibility, accurate demand forecasting and personalized marketing.

2. Social Media Fraud Detection Problem

A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

Answer:
A distributed streaming architecture is appropriate because social-media activity is continuous and extremely high volume.

• Collect posts, comments, likes, shares, account details, device signals and interaction information.
• Use Apache Kafka for high-throughput event ingestion.
• Use Apache Flink or Spark Structured Streaming for near-real-time processing.
• Calculate features such as posting frequency, account age, repeated content, unusual interactions and rapid sharing.
• Store historical activity in a distributed data lake and user features in a NoSQL database.
• Use Logistic Regression, Random Forest, Gradient Boosting or neural networks for classification.
• Use anomaly detection to identify new fraud patterns that were not present in labelled training data.
• Apply graph analytics to detect coordinated fake-account networks and suspicious content-sharing communities.
• Generate risk scores and trigger verification, human review, reduced distribution or blocking.
• Monitor precision, recall, false positives, false negatives and model drift.

Stream processing enables rapid detection, while machine learning identifies complex behavioural patterns. Distributed processing allows the system to scale by adding nodes.

3. Government Data Platform Problem

A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

Answer:
The platform should use centralized governance with distributed data ownership and standardized interfaces.

• Create a data catalogue containing dataset descriptions, owners, formats, definitions, quality information and permitted uses.
• Establish common schemas, metadata standards, identifiers and data formats.
• Use a data lake/platform to ingest census, economic and geospatial datasets through ETL/ELT pipelines.
• Apply data validation, cleansing and transformation before integration.
• Use master data management for common entities such as geographic areas and administrative units.
• Provide standardized APIs for authorized departments.
• Apply data minimization, purpose limitation, anonymization or pseudonymization where appropriate.
• Use strong authentication, role/attribute-based access control, encryption, audit logging and security monitoring.
• Assign data owners and stewards and define policies for quality, access, retention and sharing.
• Maintain data lineage and audit trails and comply with applicable privacy and security requirements.

This approach improves interoperability and data quality while ensuring that departments access only information they are authorized to use.

4. Banking Fraud Detection Problem

A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

Answer:
The bank needs a low-latency streaming pipeline combined with historical analytics.

1. Capture card, online-banking, ATM, mobile-payment, merchant, device, location and account activity.
2. Use Apache Kafka to ingest transaction events continuously.
3. Use Apache Flink or Spark Structured Streaming to calculate real-time features such as transaction velocity, amount deviation, unusual location and device changes.
4. Store historical transactions in a cloud data lake or distributed storage system.
5. Maintain frequently required features in a low-latency feature store or cache.

6. Use Logistic Regression, Decision Trees, Random Forest or Gradient Boosting for known fraud patterns.
7. Use anomaly-detection models for previously unseen behaviour.
8. Generate a real-time fraud-risk score for each transaction.
9. Approve low-risk transactions, request additional authentication for medium-risk transactions and block/investigate high-risk transactions.
10. Monitor precision, recall, false-positive rate, false-negative rate, latency and model drift.
11. Protect financial data using encryption, strict access control and audit logs.

Streaming provides immediate detection, while historical data improves model training and long-term behavioural analysis.

5. Agriculture Smart Farming Problem

A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Answer:
A precision-agriculture platform should integrate IoT sensors, weather services, drones, satellite imagery and farm records.

• Data sources: soil moisture, temperature, humidity, pH, weather stations, irrigation systems, drones, satellite imagery and crop records.
• Use MQTT/IoT gateways for sensor collection and Kafka for high-volume event streaming.
• Store raw sensor readings, images and historical crop data in a cloud data lake.
• Use time-series or NoSQL databases for fast access to current sensor readings.
• Use Apache Spark for large-scale historical processing and streaming analytics for current field conditions.
• Apply machine learning for yield prediction, crop-stress detection, disease detection, irrigation forecasting and resource optimization.
• Use CNN-based computer vision models on drone/satellite images to identify disease, weeds, nutrient stress or crop damage.
• Provide dashboards showing crop health, soil conditions, irrigation recommendations, weather risks and yield forecasts.
• Use analytics to determine where and when irrigation or fertilizer should be applied.
• Protect farmer data through authentication, encryption, access controls, backups and suitable data-sharing policies.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded: